

***What if someone is secretly listening
and controlling your internet traffic
without you even knowing? 🤔🌐***



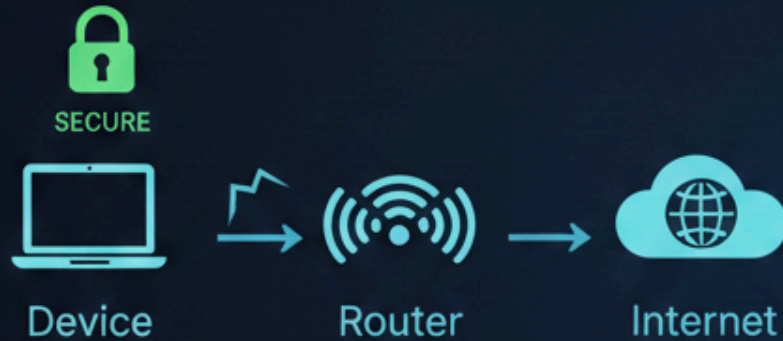
What is MITM

(Man-in-the-Middle)?

- An attack where a hacker sits between two communicating parties
- Victim believes they are talking to the router or server
- Router believes it is talking to the victim
- Attacker can observe or manipulate data

Normal Communication VS MITM Flow

🔒 Normal Flow 🦺 Communication



Device → Router → Internet

MITM Flow 🦹 Communication



Device → Attacker → Router → Internet

Attacker stays hidden in the middle



ARP Protocol

- ARP maps IP addresses to MAC addresses
- Used inside local networks
- Devices trust ARP responses automatically
- ARP has no built-in authentication



ARP Poisoning / Spoofing

- Attacker sends fake ARP replies
- Victim thinks attacker is the router
- Router thinks attacker is the victim
- This allows the attacker to become the middle point

Passive vs Active MITM

Passive MITM:

- Attacker only listens to traffic
- No data modification

Active MITM:

- Attacker modifies or injects data
- Can redirect or manipulate communication



HTTP vs HTTPS in MITM

HTTP:

- Data is sent in plain text
- Easy to read during MITM

HTTPS:

- Data is encrypted
- Much harder to read or modify
- Encryption reduces MITM damage

Real-World MITM Scenarios

- Public Wi-Fi networks (cafes, airports)
- Fake or open Wi-Fi hotspots
- Compromised office networks
- Shared home networks

Real-Life Impact of MITM

- Login credentials can be stolen
- Session hijacking can occur
- Sensitive data can be modified
- Financial and identity theft risks



Detection & Prevention

How to Prevent MITM?

- Always use HTTPS websites
- Watch for certificate warnings
- Avoid open public Wi-Fi
- Use VPN on untrusted networks
- Keep systems and routers updated